

# Quantum theory of radiation for nonstationary systems

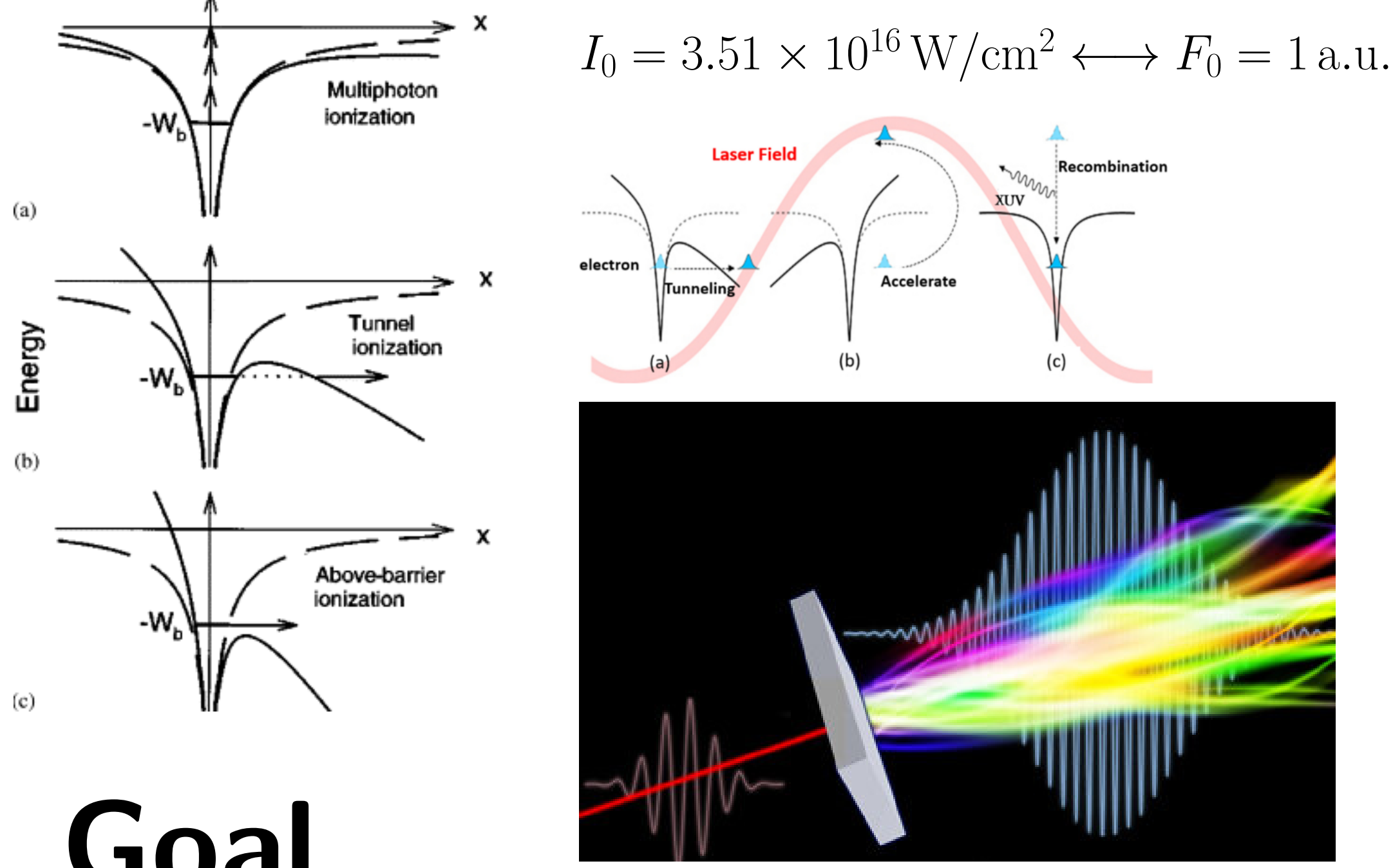
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High harmonic generation (HHG) and photoelectron momentum distribution (PEMD) are some of the most important processes driving the study of matter-radiation interaction and will lead to important advances in fundamental science and engineering. Through HHG, for example, it is possible to produce coherent radiations outperforming the external fields in regards to frequency and pulse duration. The current theory, valid for stationary systems, falls short at the description of nonstationary systems, since it ignores radiations induced by the external force. Following Yangeliev *et al.*[1], we aim at extending the quantum theory of radiation for real nonstationary systems, i.e in 3 dimensions and for realistic potentials.

## 1. Background

Basic strong field phenomena



## 2. Goal

Full quantum treatment of the HHG theory

Schrödinger equation

$$i\frac{\partial |\Psi(t)\rangle}{\partial t} = \left[ \hat{H}_a + \hat{H}_f + \alpha \hat{\mathbf{A}} \hat{\mathbf{p}} + \hat{V}_{\text{ext}}(t) \right] |\Psi(t)\rangle,$$

$$\hat{H}_a = \frac{\hat{\mathbf{p}}^2}{2} + \hat{V}_a \quad |\Psi(t)\rangle = \sum_n A_n(t) e^{-iE_n t} |\psi_n^-, 0\rangle$$

$$\hat{H}_a |\psi_n^\pm\rangle = E_n |\psi_n^\pm\rangle \quad + \alpha \sum_{n\nu} B_{n\nu}(t) e^{-i(E_n + \omega)t} |\psi_n^-, \nu\rangle.$$

$$\hat{H}_f = \sum_\nu \omega \hat{a}_\nu^\dagger \hat{a}_\nu, \quad \hat{H}_f |\nu\rangle = \omega |\nu\rangle$$

$$\hat{R}_{\nu} = \int e^{i\hat{H}_a t} \hat{\mathbf{A}}_\nu \hat{\mathbf{p}} e^{-i\hat{H}_a t} e^{i\omega t - \alpha t} dt$$

$$|\psi_n^-, 0\rangle = |\psi_n^-\rangle |0\rangle + i\alpha \sum_\nu \hat{R}_\nu |\psi_n^-\rangle |\nu\rangle + O(\alpha^2),$$

$$|\psi_n^-, \nu\rangle = |\psi_n^-\rangle |\nu\rangle - i\alpha \hat{R}_\nu^\dagger |\psi_n^-\rangle |\nu\rangle + \alpha \times (\text{two-photons states}) + O(\alpha^2).$$

PEMD and HHG in a full quantum treatment

$$P_n = |\langle \psi_n^-, 0 | \Psi(t \geq T) \rangle|^2, \quad Q_\nu = \sum_n |\langle \psi_n^-, \nu | \Psi(t \geq T) \rangle|^2.$$

These observables are usually evaluated at the order 0 of the perturbation theory ( $\alpha \rightarrow 0$ ), between some more approximations

## 3. Illustration

The classical ansatz (semi-classical treatment)

$$i\frac{\partial \psi(x, t)}{\partial t} = \left[ -\frac{1}{2} \frac{d}{dx^2} - \kappa \delta(x) + F(t)x \right] \psi(x, t),$$

$$i\frac{\partial \phi_\omega(x, t)}{\partial t} = \left[ -\frac{1}{2} \frac{d}{dx^2} - \kappa \delta(x) + F(t)x \right] \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t),$$

$$\phi_\omega(x, 0) = \frac{-i\kappa^{3/2}}{\omega} \text{sgn}(x) (e^{-\kappa|x|} - e^{-\kappa(\omega)|x|}), \quad \psi(x, 0) = \kappa^{1/2} e^{-\kappa|x|}, \quad E_0 = -\kappa^2/2.$$

$$a_0 = e^{iE_k t} \int \psi(x, 0) \psi(x, T) dx, \quad a_k = e^{iE_k t} \int \psi_k^{(-)*}(x) \psi(x, T) dx,$$

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'} dk'}{E_0 + \omega - E_{k'} + i0/2\pi},$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_0 + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'} dk'}{E_k + \omega - E_{k'} + i0/2\pi} r.$$

Convergence assessment observables

$$\rho(x, t) = |\psi(x, t)|^2, \quad P_0 = |a_0|^2, \quad P_0 + P_{\text{ion}} = 1.$$

Alternative PEMD and HHG expressions in the classical ansatz[1]

$$P(k) = |a_k|^2, \quad Q_\nu^{(c)} = \alpha^2 \omega^2 |\mathbf{A}_\nu \mathbf{d}(\omega)|^2$$

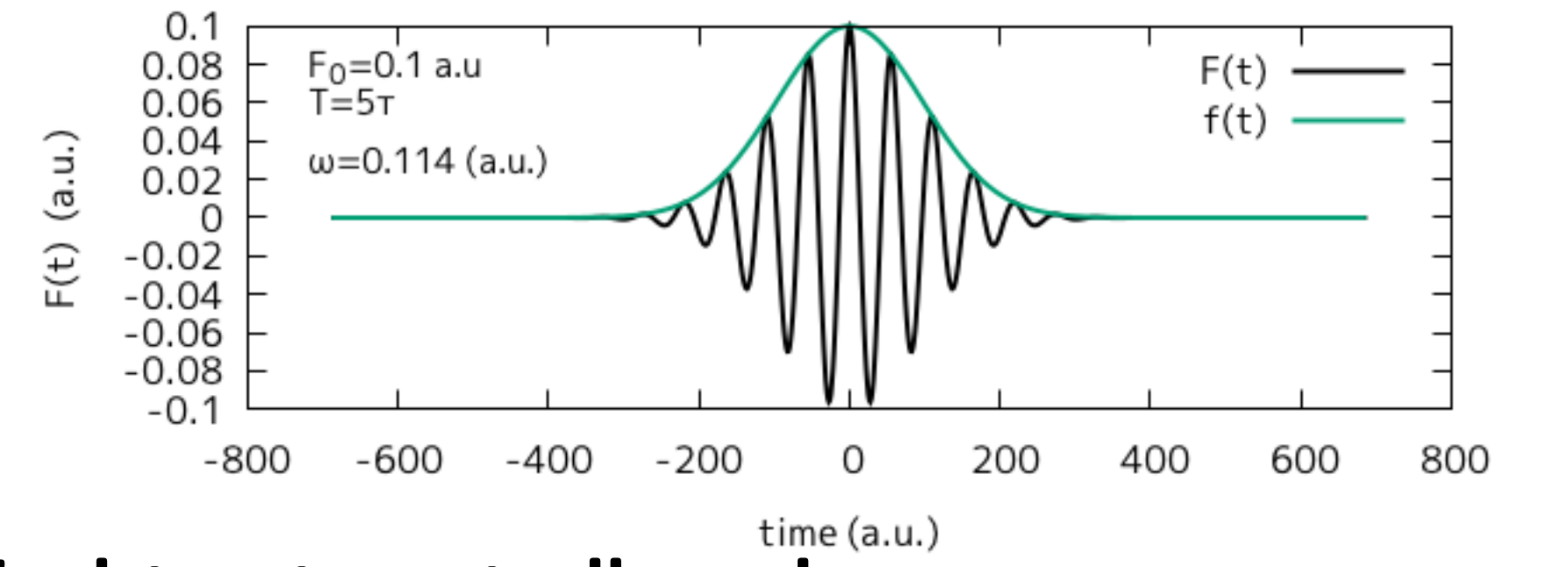
Quantum Field

$$\mathbf{A} = \sum_\nu \mathbf{A}_\nu (\hat{a}_\nu^\dagger + \hat{a}_\nu),$$

$$\mathbf{F}(t) = -\frac{\partial \mathbf{A}(t)}{\partial t}$$

If  $F_0 \gg 0.005 \text{ a.u.} \rightarrow$  Classical treatment allowed

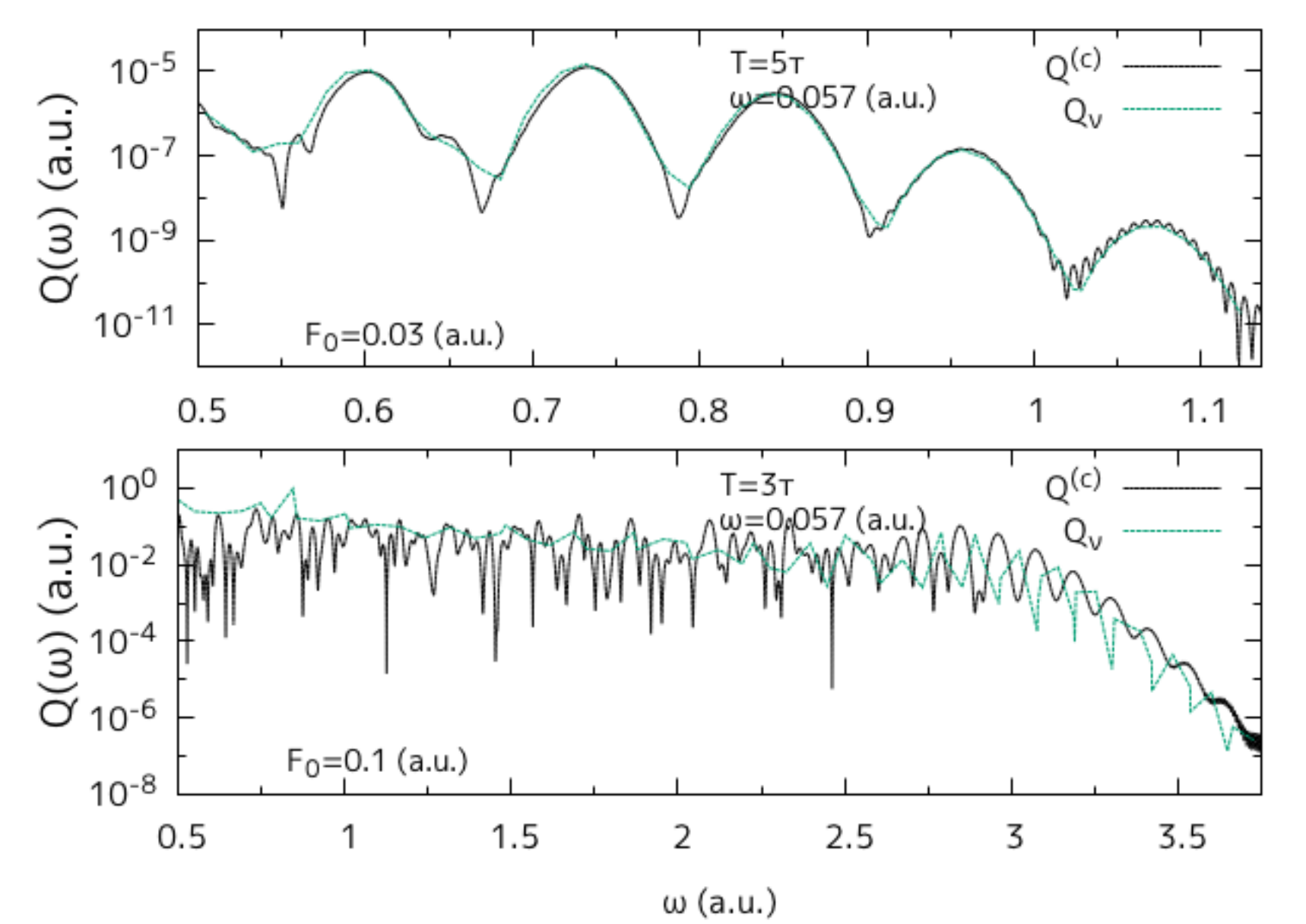
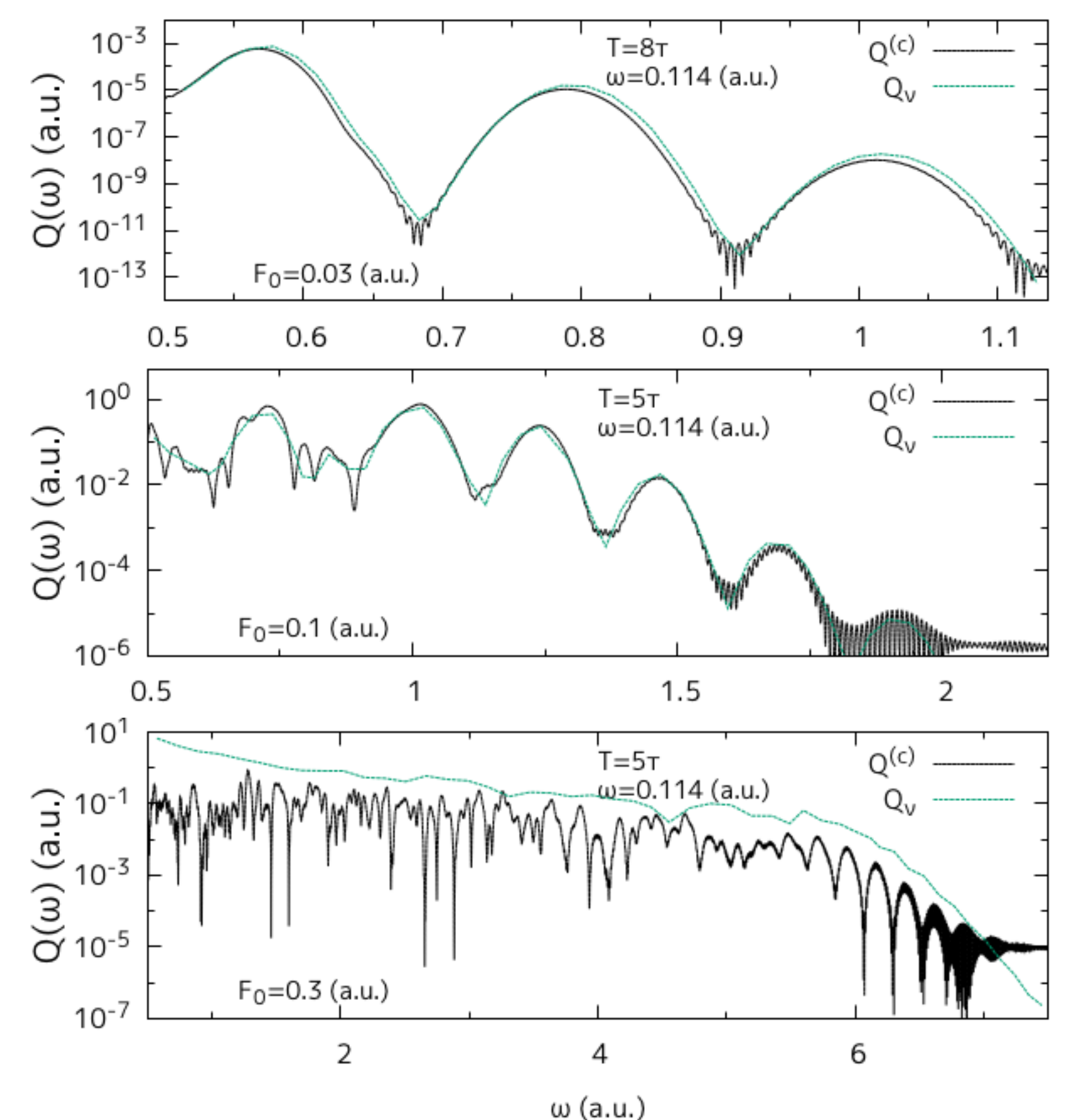
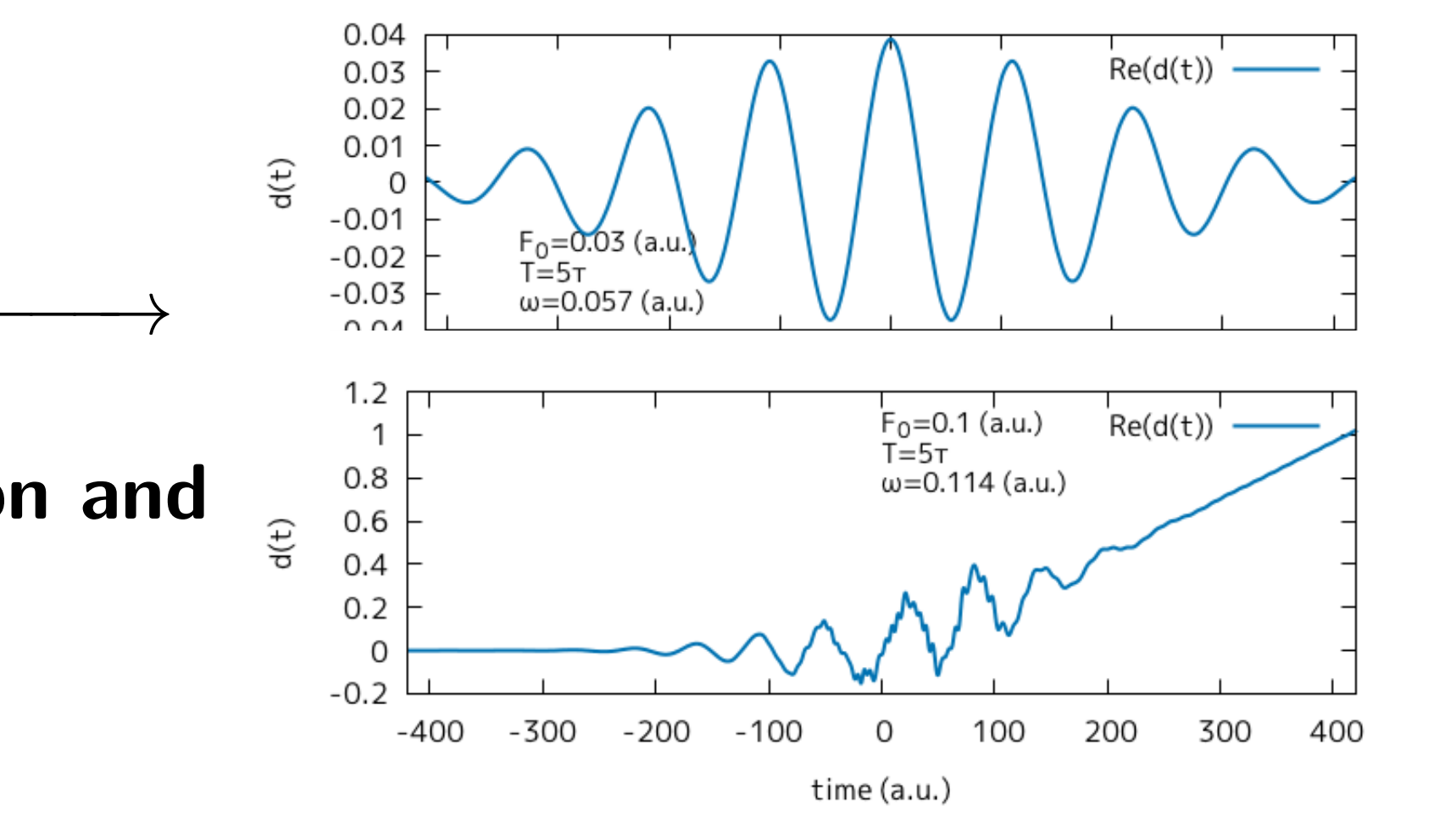
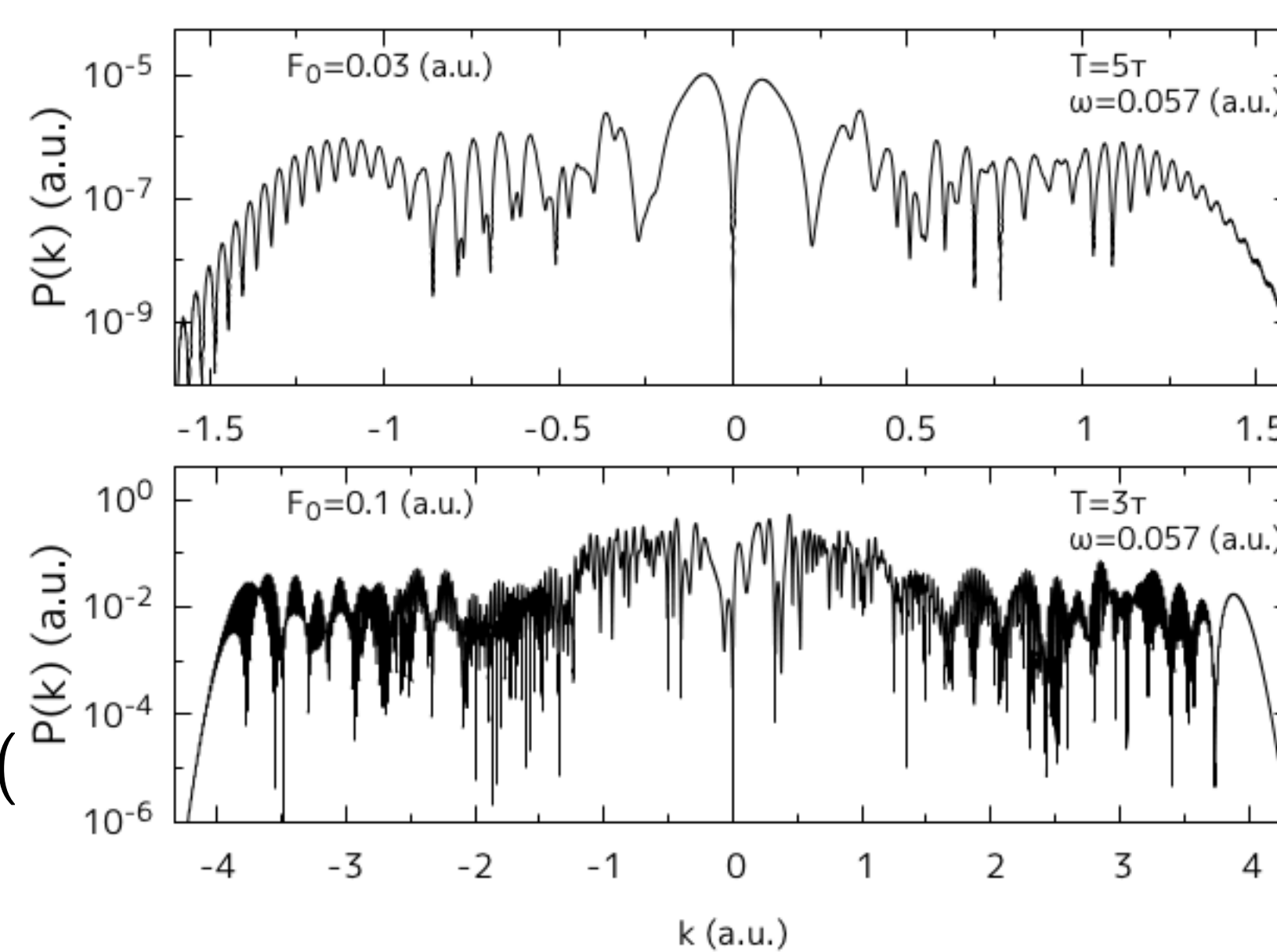
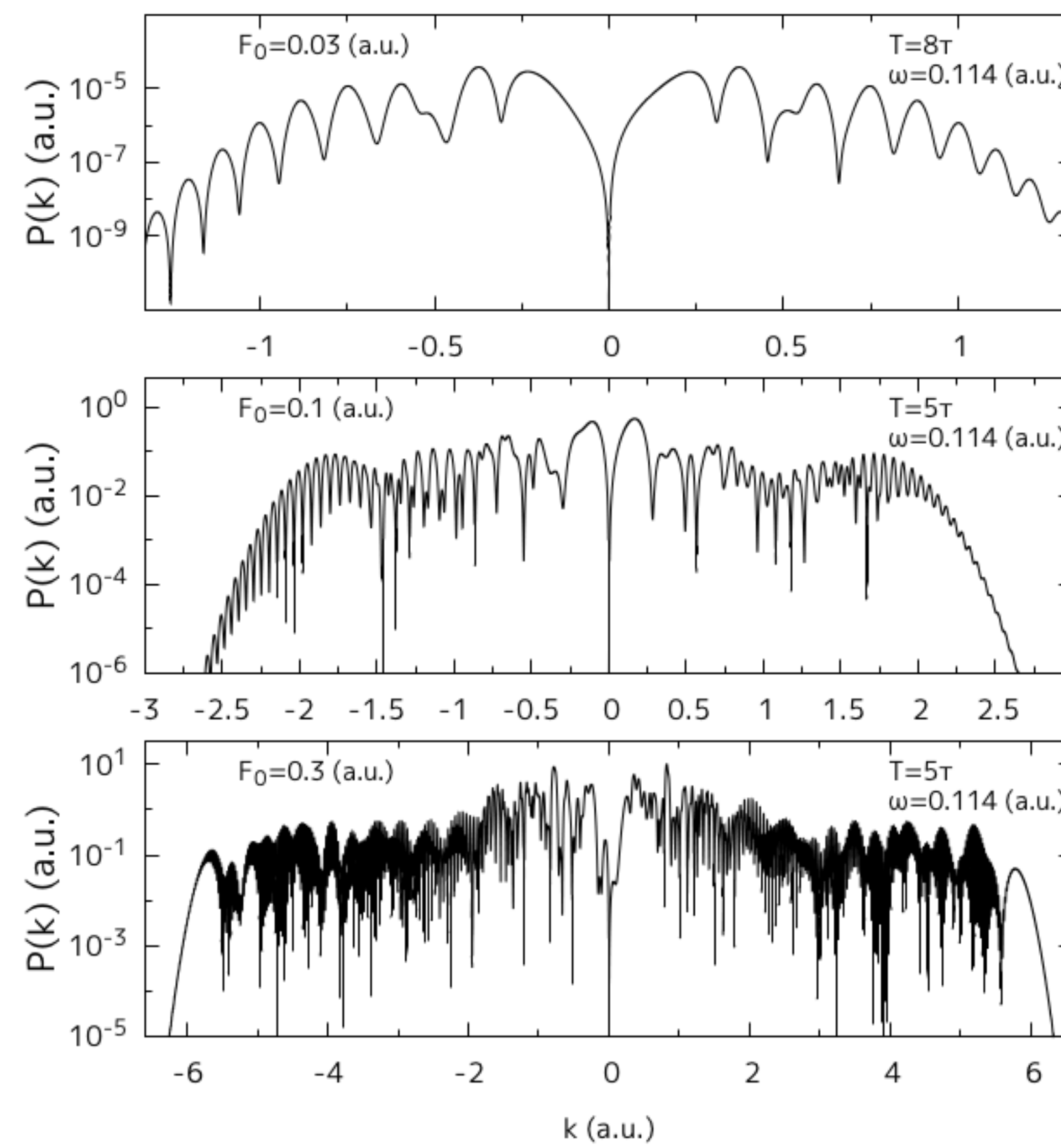
$$\mathbf{F}(t) = \mathbf{F}_0 e^{-(2t'/\tau)^2} \cos(\omega t'), \quad t' = t - T/2, \quad 0 \leq t \leq T, \quad \tau = 2\pi n_{\text{oc}}/\omega, \quad n_{\text{oc}} = 5, \quad T = n\tau$$



## 4. Results

Dipole moment of the particle

Photoelectron momentum distribution and High harmonic generation



## 5. Conclusion

- The classical ansatz used to evaluate HHG spectra ceases to be a good approximation for field strengths approaching unity
- Since lasers of intensity greater than  $I_0$  are now available, a more appropriate theoretical treatment for HHG is highly desirable
- The present study aims at extending this previous result to realistic atomic system

## References

- [1] D. N. Yangeliev, V. P. Krainov and O. I. Tolstikhin, Phys. Rev. A 101, 013410 (2020) [3] M. Ohmi, O. I. Tolstikhin, and T. Morishita, Phys. Rev. A 86, 013412 (2015) [5] O. I. Tolstikhin, T. Morishita, and S. Watanabe, Phys. Rev. A 81, 033415 (2010)
- [2] O. I. Tolstikhin, T. Morishita, and S. Watanabe, Phys. Rev. A 81, 033415 (2010) [4] D. Kidd, C. Covington, and K. Varga, Phys. Rev. E 96, 063307 (2017) [6] W. Heitler, *The Quantum theory of Radiation* (Dover, nEw York, 1984)